

Pi Good-Cop / Bad-Cop Workflow

Lean dev loop for Pi v0.85.1 on Opencode Zen free models - Debian 12, 6.3 Gi RAM

<https://pi.dev> | *sequential* | *no bloat* | *Sep 10, 2026*

Builder (good cop) makes it pass. Critic (bad cop) tries to break it. You stay judge. This README documents the lean, sequential loop built for your Pi setup: v0.85.1, Opencode Zen free models, 6.3 Gi RAM. No sub-agents, no parallel bloat - one pi -p call at a time.

Figure 1 - Pi v0.85.1 startup in ~/d - muse-spark-1.2 at high thinking - 28 skills, 14 prompts loaded

```
pi v0.85.1
escape interrupt · ctrl+c/ctrl+d clear/exit · / commands · ! bash · ctrl+o more
Press ctrl+o to show full startup help and loaded resources.

Pi can explain its own features and look up its docs. Ask it how to use or extend Pi.

[Context]
~/pi/agent/AGENTS.md

[Skills]
brainstorming, code-review, codebase-design, context-keeper, diagnosing-bugs, domain-modeling, executing-plans,
finishing-a-development-branch, grill-with-docs, grilling, handoff, hf-cli, implement, improve-codebase-architecture,
karpathy-guidelines, prototype, receiving-code-review, research, systematic-debugging, test-driven-development, to-spec,
to-tickets, using-git-worktrees, using-superpowers, verification-before-completion, wayfinder, wizard, writing-for-agents,
writing-plans, writing-skills

[Prompts]
/changelog, /cleanup, /commit, /docs, /explain, /find-bugs, /fix, /onboard, /plan, /pr, /refactor, /review, /simplify,
/sync-free-models, /test

~/d
0.0%/1.0M (auto) muse-spark-1.2-contributor-free · high
[alpha] 0:top 1:e 2:agy 3:pi* 4:sh- 5:git 6:ol 7:msc /home/rgroot/d 1.6% Sep-10 09:54
```

Screenshot: Screenshot 2026-09-10 9.55.00 AM.png - full startup help and loaded resources

Why 2 Models?

- Diversity: free models have different strengths. Pairing builder + critic catches blind spots.
- Low RAM: sequential pi -p keeps memory < 300 MB. No parallel agents.
- Free tier safe: 8 s sleep between calls avoids 429 rate limits.
- You decide: critic lists MUST-FIX vs NICE-TO-HAVE. If LGTM, fixer step skips.

Models in this repo (from settings.json)

Role	Model	Thinking	Strength
Builder	muse-spark-1.3-free	high	code, TDD
Critic	big-pickle	high	reasoning
Fixer	muse-spark-1.2-free	high	small fixes

Override without editing: BUILDER=opencode/mimo-v2.5-free:high CRITIC=opencode/nemotron-3-ultra-free:high ./goodcop-badcop.sh "task"

Quick Start

```
chmod +x goodcop-badcop.sh

# single task
./goodcop-badcop.sh "implement task 1 from @docs/plans/2026-09-10-auth.md - TDD"

# with file context
./goodcop-badcop.sh "fix bug in @src/core.clj - add edge case test"

# custom models
BUILDER=opencode/mimo-v2.5-free:high \
CRITIC=opencode/nemotron-3-ultra-free:high ./goodcop-badcop.sh "my task"
```

Output: .pi-runs/YYYY-MM-DD_HHMMSS/ with 1-build.md, 2-critic.md, 3-fix.md. Inspect, then continue:

```
cat .pi-runs/*/2-critic.md
pi -c # back to interactive, full history
```

Loop Over a Whole Plan

```
for i in 1 2 3; do
  ./goodcop-badcop.sh "implement task $i from @docs/plans/2026-09-10-auth.md - TDD, run clojure.test after"
done
```

Recommended Pi Workflow (6 Phases)

- 1. Start session: pi --name "feature-x" or pi -c to resume. Name early for pi -r search.
 - 2. Design: /skill:brainstorming -> /skill:writing-plans -> docs/plans/*.md (wayfinder if huge).
 - 3. Isolate: /skill:using-git-worktrees - one branch per plan, never on main.
 - 4. Build loop: /skill:test-driven-development - bb for speed, CIDER REPL live, clojure.test + clj-kondo + cljfmt.
 - 5. Branch & recover: Enter=steering, Alt+Enter=follow-up, /tree alt path, /clone new file, /compact when long.
 - 6. Verify & finish: /skill:verification-before-completion -> /skill:code-review -> /skill:finishing-a-development-branch.
- Skill map: research (facts), diagnosing-bugs (hunt), codebase-design (deepen), wizard (manual setup), context-keeper (long tasks).

Pi Session Tips

- @ to attach files, Tab to complete path, Shift+Enter multi-line, !cmd to LLM, !!cmd silent
- Ctrl+P cycle free models, Shift+Tab cycle thinking, Ctrl+O collapse tools, Esc Esc = /tree
- /reload after editing skills or AGENTS.md. /export or /share to save session HTML.
- Ctrl+G opens \$EDITOR (set to emacs for CIDER). Ctrl+L picks model, Ctrl+S saves default.

Files in This Repo

File	Purpose
goodcop-badcop.sh	3-step loop: build -> critic -> fix (env-overridable)
Screenshot 2026-...	Pi startup reference (Fig. 1)
README.md	This doc - markdown
README.pdf	Printable - with embedded screenshot

Lean note: prefer bb over cold JVM, keep deps.edn minimal, avoid large npm installs on 11 GB free disk. Ask before large downloads.

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